

Puppala Madhuvenu

Java Full Stack Developer — Java — Spring Boot

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Professional Summary

B.Tech Computer Science Graduate (2026) with **CGPA 8.99/10** and hands-on software development experience using **Java, Spring Boot, Hibernate/JPA, REST APIs, MySQL, JavaScript, and React.js**. Experienced in **object-oriented design, data structures, API development, debugging, database optimization, and performance analysis**. Built full-stack and backend applications involving **Spring Security, WebSocket, and concurrent load testing**, including measurable improvements in API latency and throughput. Strong foundation in software engineering, problem solving, and maintainable application development.

Education

PBR Visvodaya Institute of Technology and Science , Kavali	2022 – 2026
B.Tech in Computer Science and Engineering	CGPA: 8.99/10
Sai Cooperative Junior College , Kavali	2020 – 2022
Intermediate (MPC)	70.60%
AP Model Government School , Kondapuram	2019 – 2020
SSC	89.33%

Skills

Programming Languages: Java, Python, JavaScript, SQL
Backend: Spring Boot, Spring Security, JPA/Hibernate, RESTful APIs
Web Technologies: HTML5, CSS3, React.js, WebSocket
Software Engineering: OOP, Data Structures & Algorithms, SDLC, Debugging, Modular Architecture
Databases: MySQL, Relational Database Design, SQL Query Optimization
Tools: Git, GitHub, Maven, Postman, VS Code

Projects

AI-Powered Intelligent Smart Queue Management System	Live Demo	GitHub	May 2026 – Jul 2026
- Architected a real-time queue management platform using Java, Spring Boot, MySQL, REST APIs, WebSocket, and modular architecture, separating appointment scheduling, token generation, real-time queue management, ETA recalculation, and notification domains. - Optimized Spring Security authentication by eliminating redundant database lookups and reusing authenticated user context, reducing login latency by 44% (498 ms → 280 ms), p95 latency by 48% (1.52 s → 793 ms), and increasing throughput by 17% (30 → 35 req/s) under 100 concurrent users using k6. - Validated backend across authentication, appointment booking, queue management, analytics, and doctor workflows, sustaining 43.7 req/s, 26 ms average latency, 98.6 ms p95 latency, and 0% request failures under 100 concurrent users using k6. - Reduced analytics API p95 latency by 45% (1.30 s → 711 ms) and average latency by 58% (419 ms → 177 ms) at 1,000 concurrent virtual users by eliminating N+1 database access and unnecessary entity loading while maintaining 0% request failures.			
Enterprise Employee Management Platform – Java Spring Boot	Live Demo	GitHub	Jan 2025 – Mar 2025
- Developed a full-stack employee administration platform using Java, Spring Boot, Spring Security, Hibernate/JPA, MySQL, REST APIs, and React.js, implementing employee, department, document, and dashboard workflows. - Implemented JWT-based stateless authentication and role-based authorization for ADMIN, HR, and MANAGER users using Spring Security, including access and refresh token handling and protected REST endpoints. - Built scalable employee APIs with server-side pagination, filtering, sorting, JPA Specifications, validation, global exception handling, and audit logging, improving maintainability and preventing inefficient retrieval of large employee datasets. - Applied JPA entity relationships, transactional service-layer operations, document/file management, and Docker-based application packaging for consistent application deployment and data management.			

Professional Experience

CodeTantra	May 2025 – Jun 2025
<i>Java Full Stack Intern</i> Certificate	
- Implemented hotel booking workflows using Java, Spring Boot, MySQL, REST APIs, and persistence logic, translating business requirements into maintainable application components. - Reduced API latency by 25% (400 ms → 300 ms) by tracing slow hotel-booking requests to inefficient database operations, optimizing the underlying SQL queries, and refining Spring Boot service-layer processing to eliminate unnecessary work.	